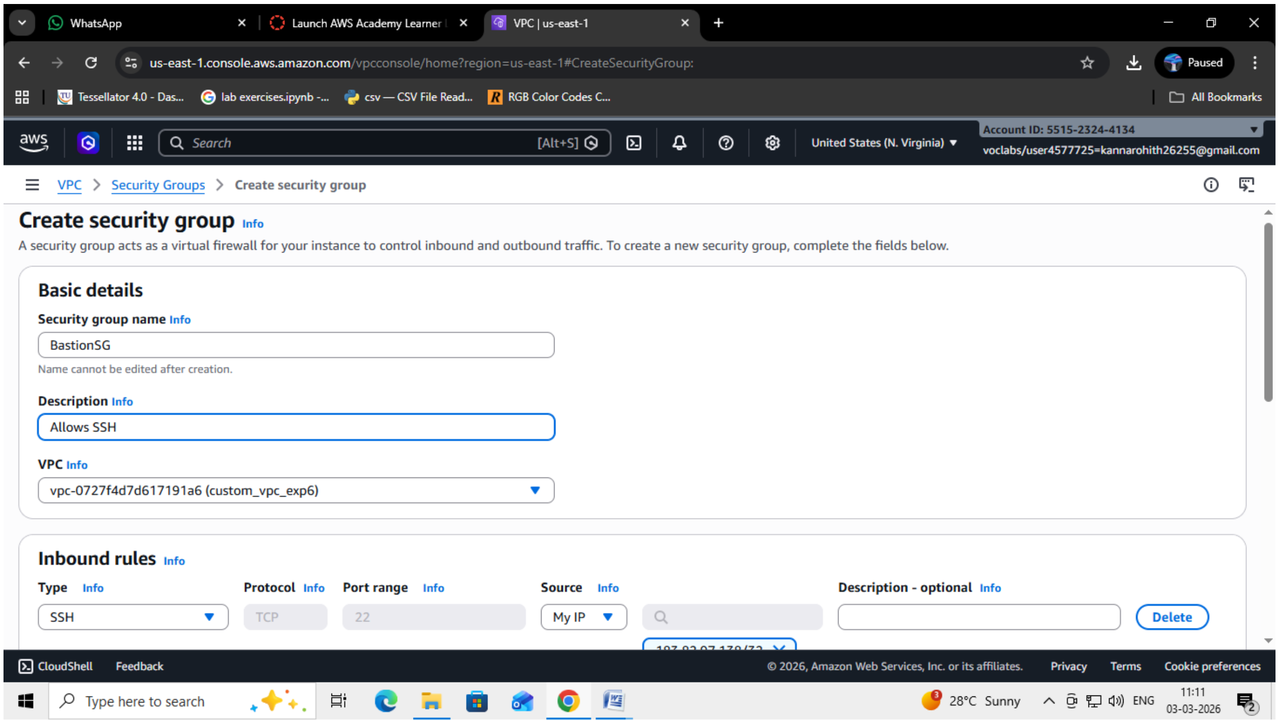
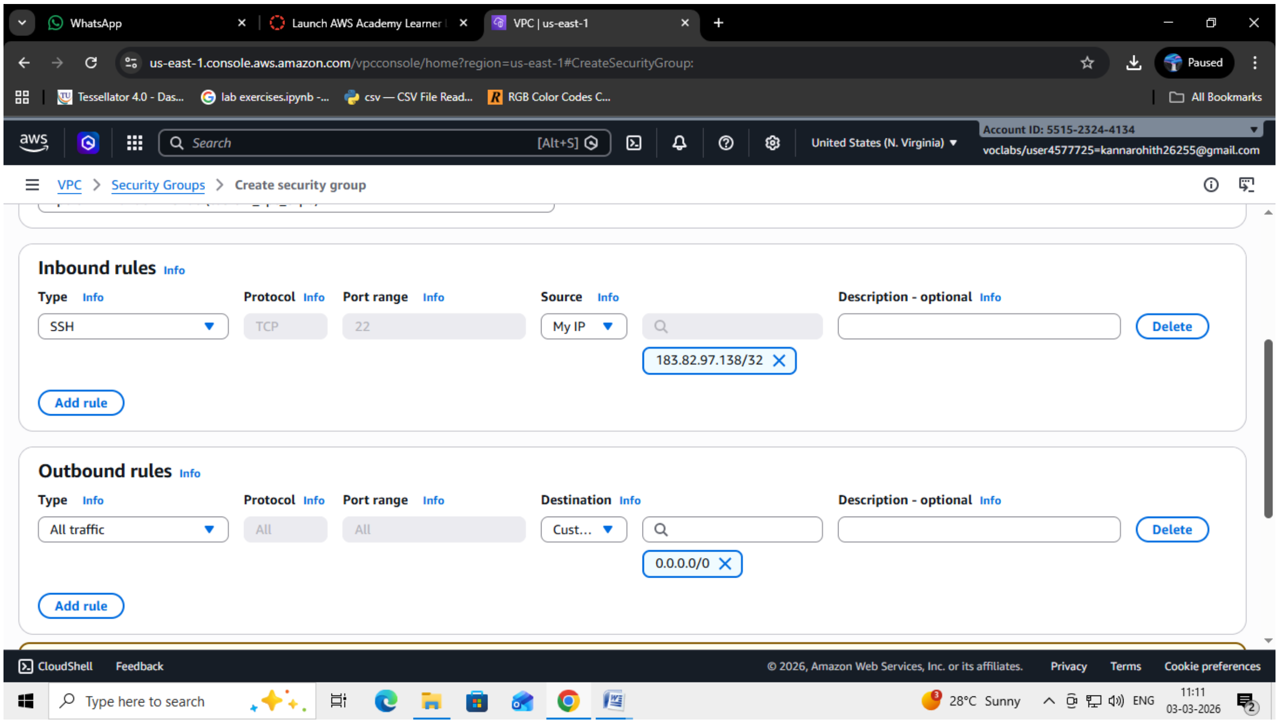
VPC USING BASTION SERVER

1. **VPC with Bastion Server**
   VPC & Subnet Design
   VPC CIDR: 10.0.0.0/16
   Public Subnet (Bastion Server): 10.0.1.0/24
   Private Subnet (Database Server): 10.0.2.0/24

---

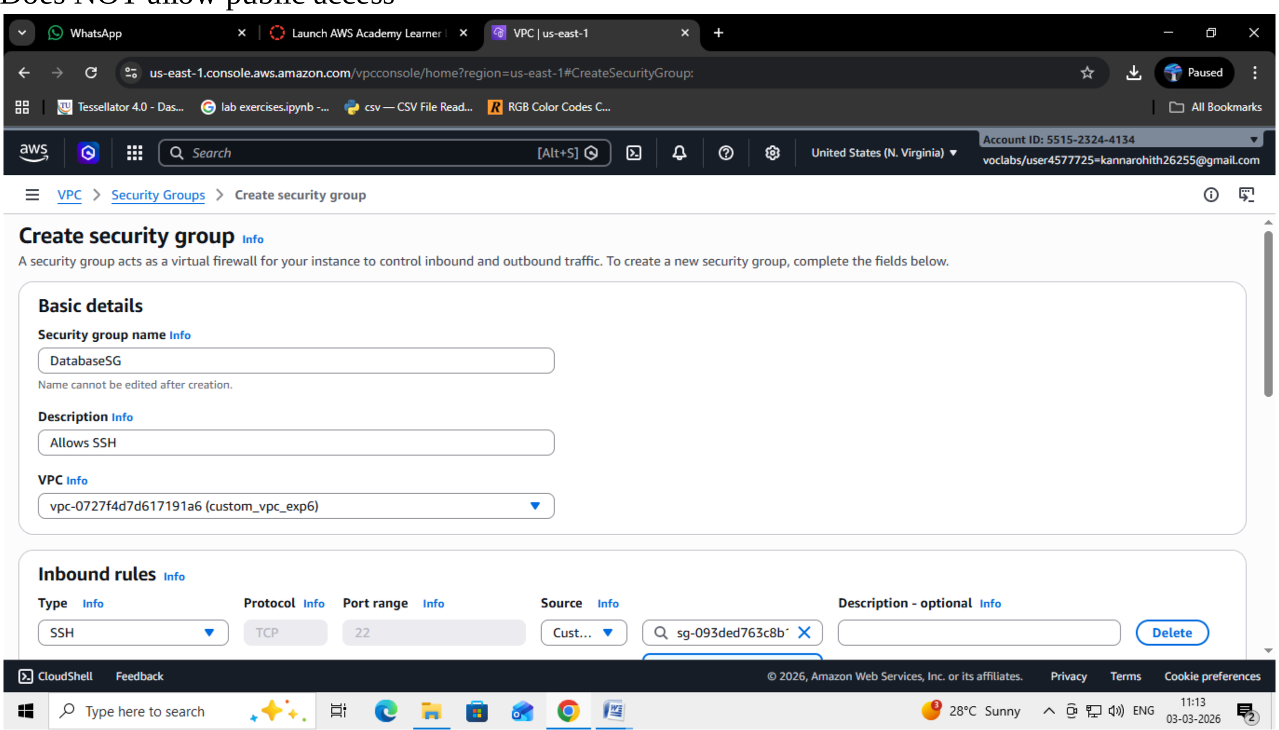
2. **Security Group Configuration (VERY IMPORTANT)**
   Bastion Server Security Group (Public Subnet)
   Allows SSH (22) from:My IP (Database Administrator)

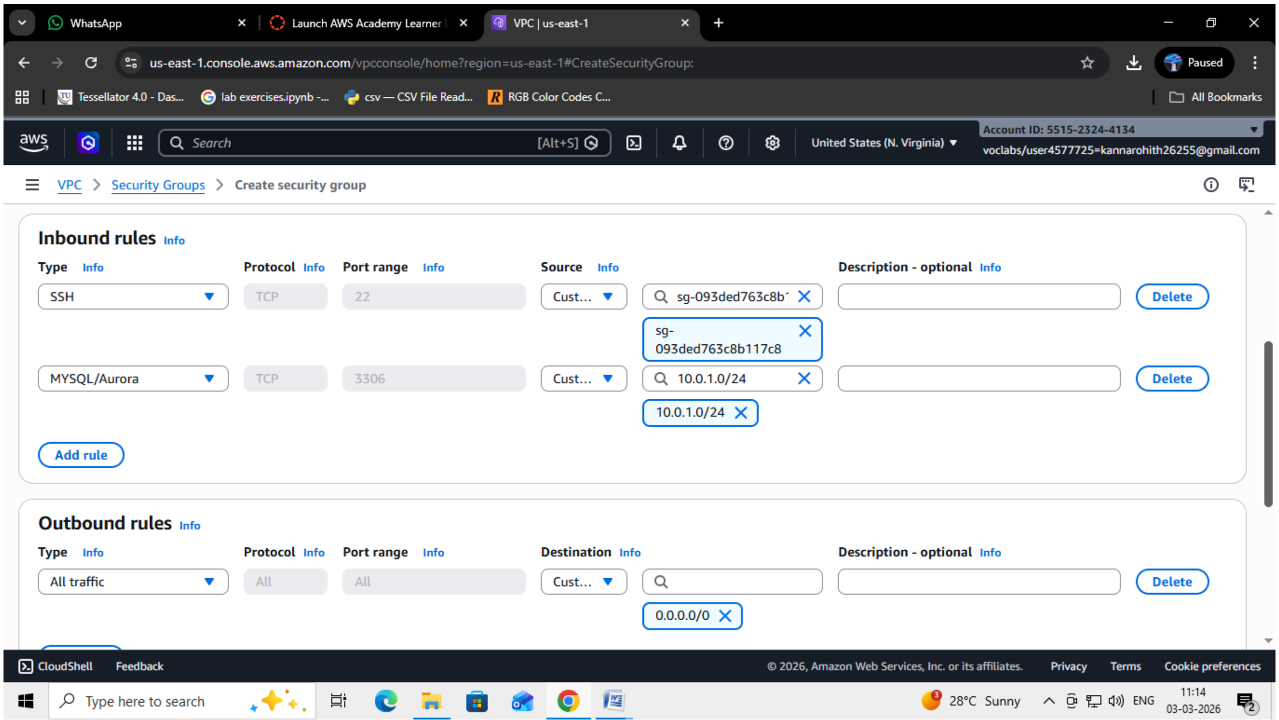


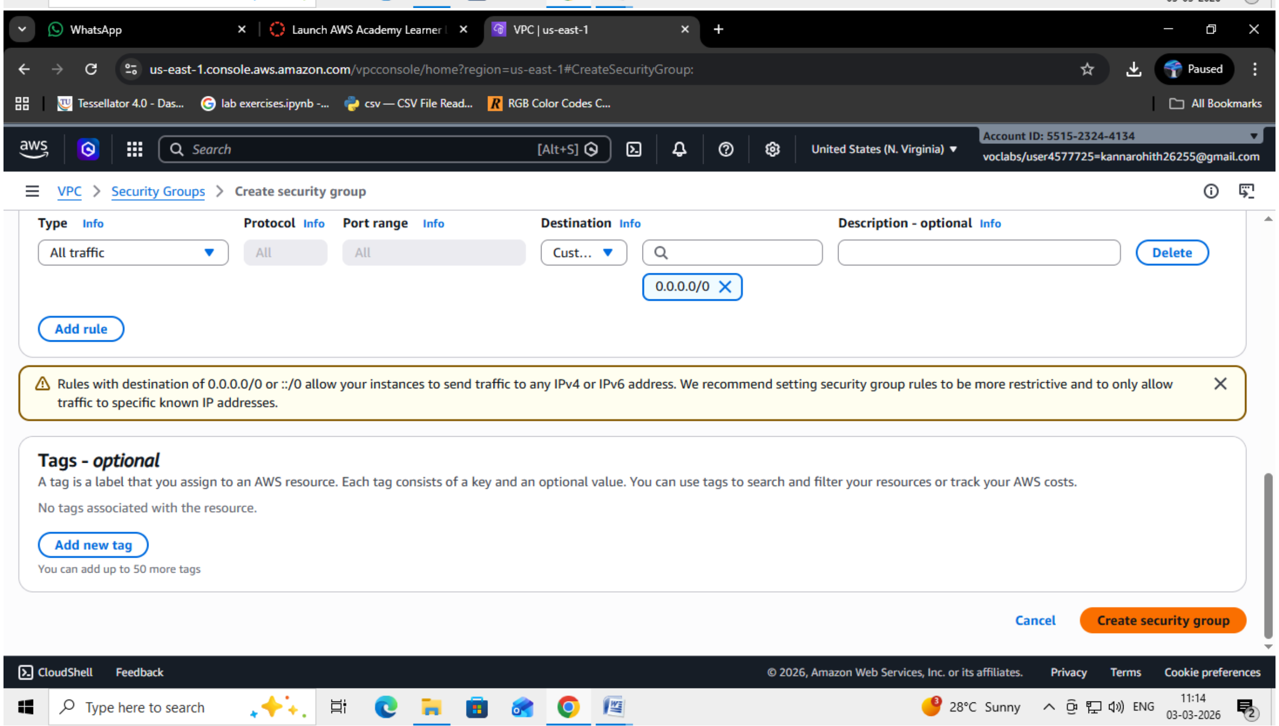


Database Server Security Group (Private Subnet)

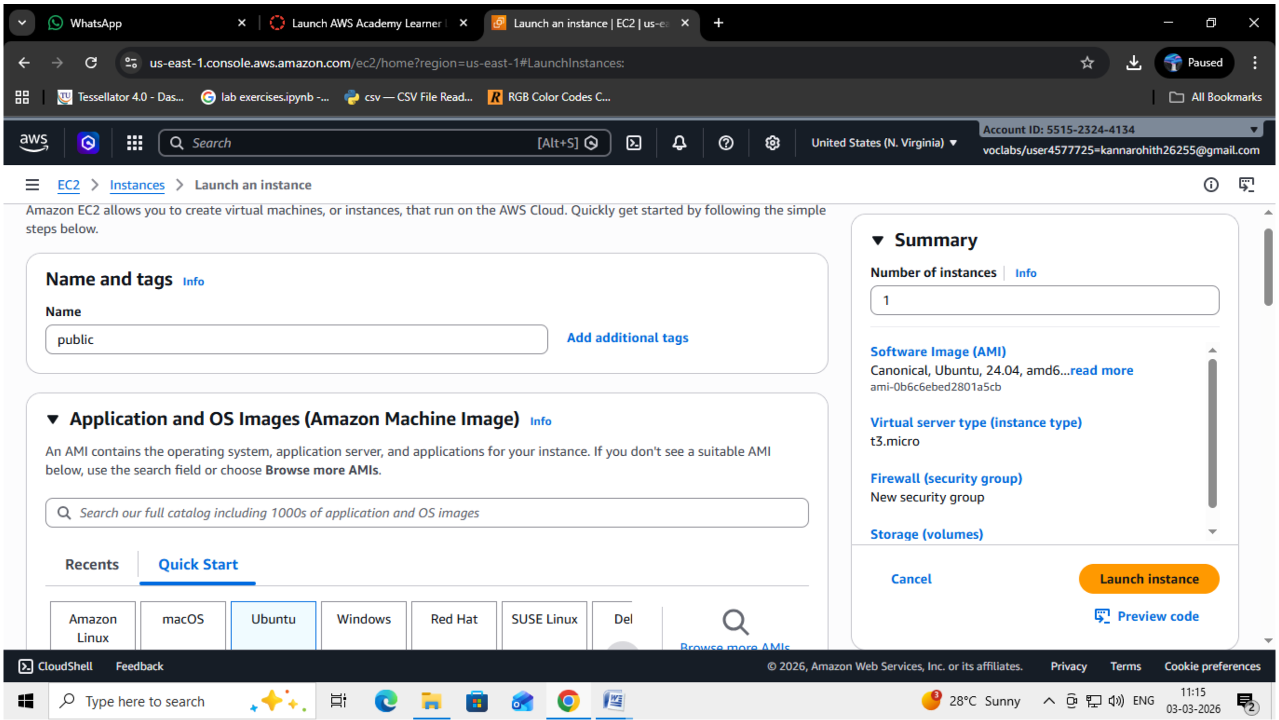
Allows MySQL / Aurora (3306) from: custom 10.0.1.0/24 (Public subnet / Bastion subnet)
Allows SSH (22) from: custom Private IP of Bastion Server
Does NOT allow public access



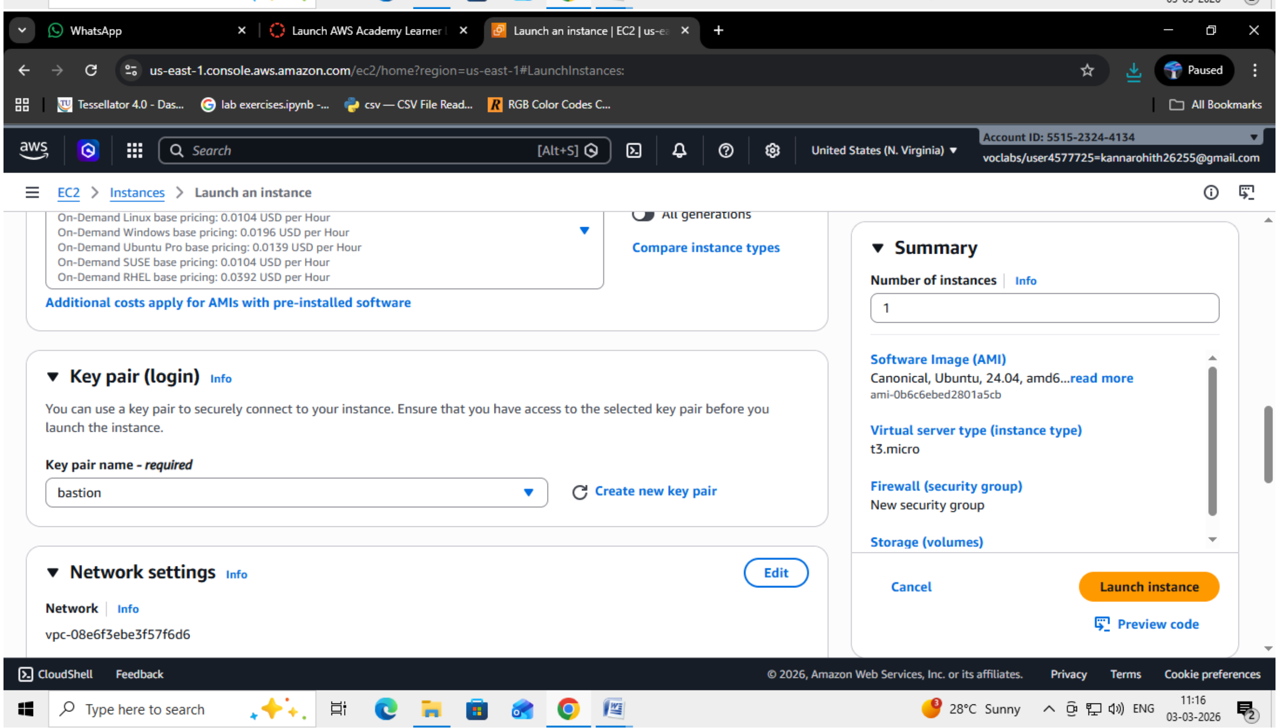




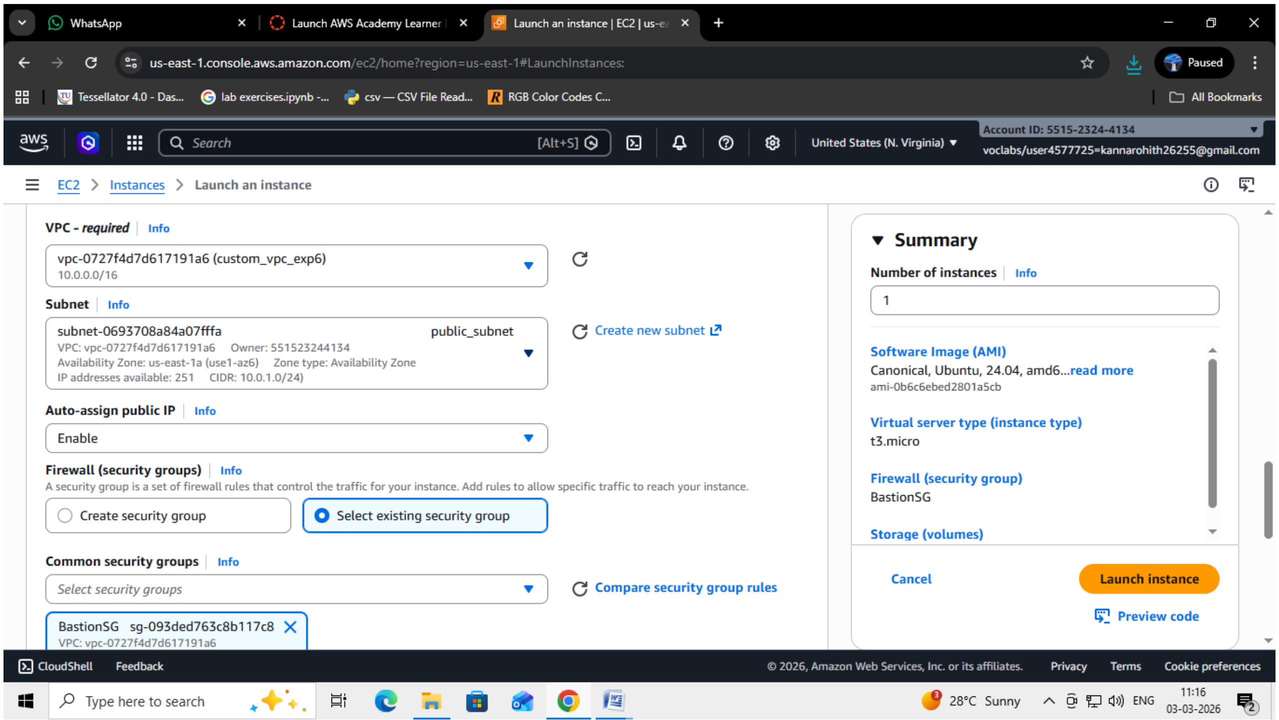
3. **Security Outcome**
   Database server is NOT publicly accessible
   Only Bastion server can connect to DB server using SSH
   External users cannot access DB directly
4. **EC2 Instance Creation**
   Bastion Server
   Subnet: Public Subnet (10.0.1.0/24)
   Key pair: bastion.pem
   Operating System: Ubuntu



WhatsApp
Launch AWS Academy Learner
Launch an instance | EC2 | us-e
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Tessellator 4.0 - Das...
lab exercises.ipynb -...
csv — CSV File Read...
RGB Color Codes C...
All Bookmarks
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5515-2324-4134
voclabs/user4577725=kannarohith26255@gmail.com
EC2 > Instances > Launch an instance
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
public
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Del
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0b6c6ebed2801a5cb
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
Cancel
Launch instance
Preview code
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
28°C Sunny
ENG
11:15
03-03-2026



WhatsApp
Launch AWS Academy Learner
Launch an instance | EC2 | us-e
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Tessellator 4.0 - Das...
lab exercises.ipynb -...
csv — CSV File Read...
RGB Color Codes C...
All Bookmarks
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5515-2324-4134
voclabs/user4577725=kannarohith26255@gmail.com
EC2 > Instances > Launch an instance
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
bastion
Create new key pair
Network settings Info
Edit
Network Info
vpc-08e6f3ebe3f57f6d6
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0b6c6ebed2801a5cb
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
Cancel
Launch instance
Preview code
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
28°C Sunny
ENG
11:16
03-03-2026

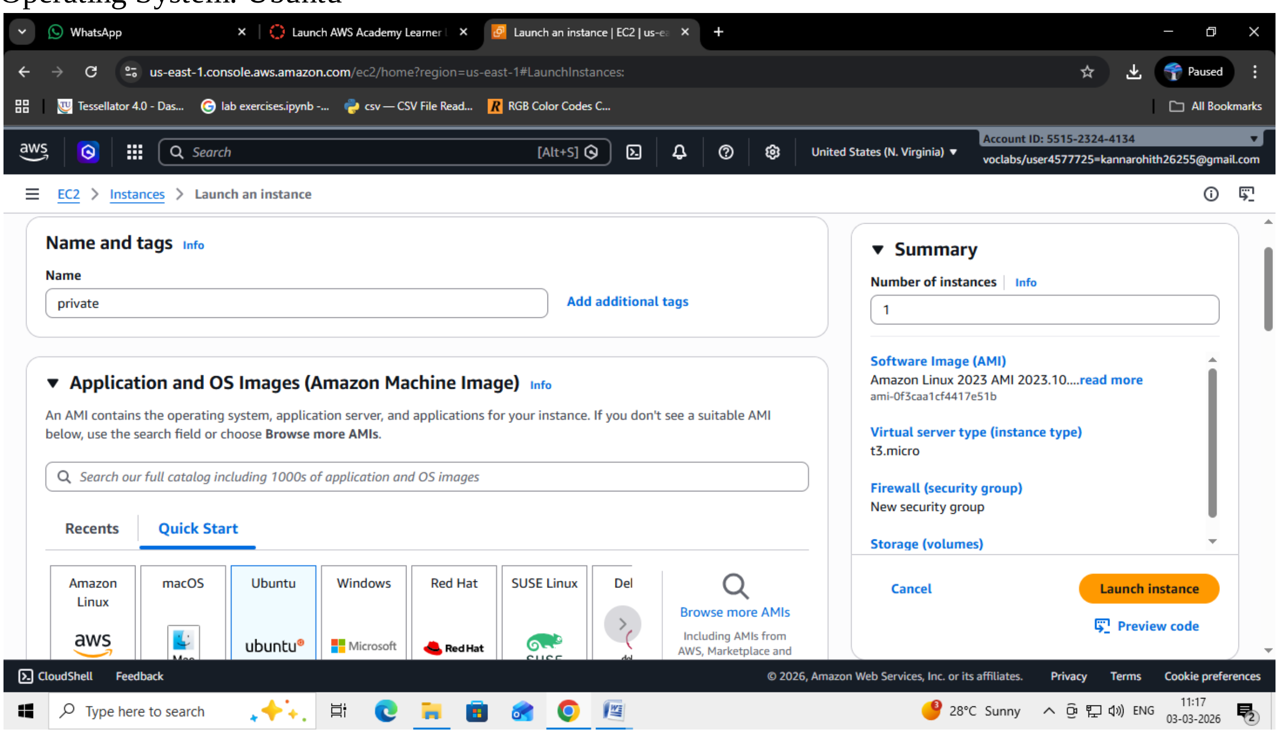


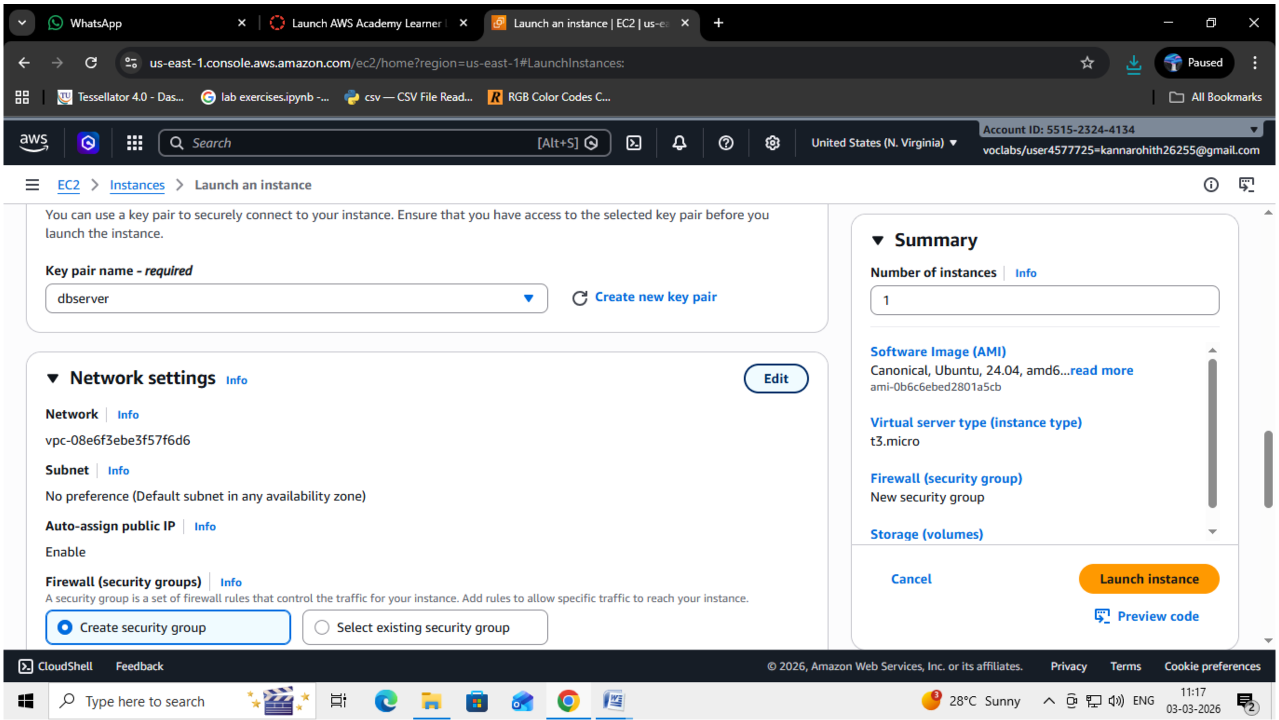
Database Server

Subnet: Private Subnet (10.0.2.0/24)

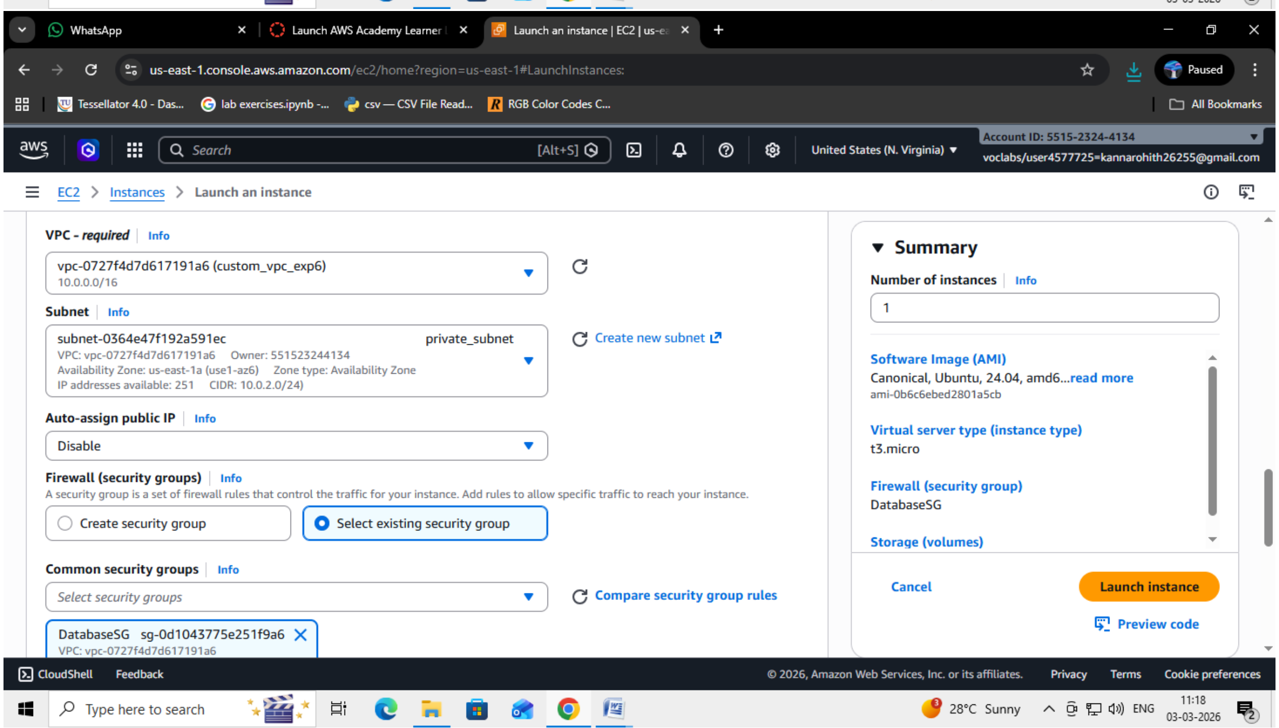
Key pair: dbserver.pem

Operating System: Ubuntu

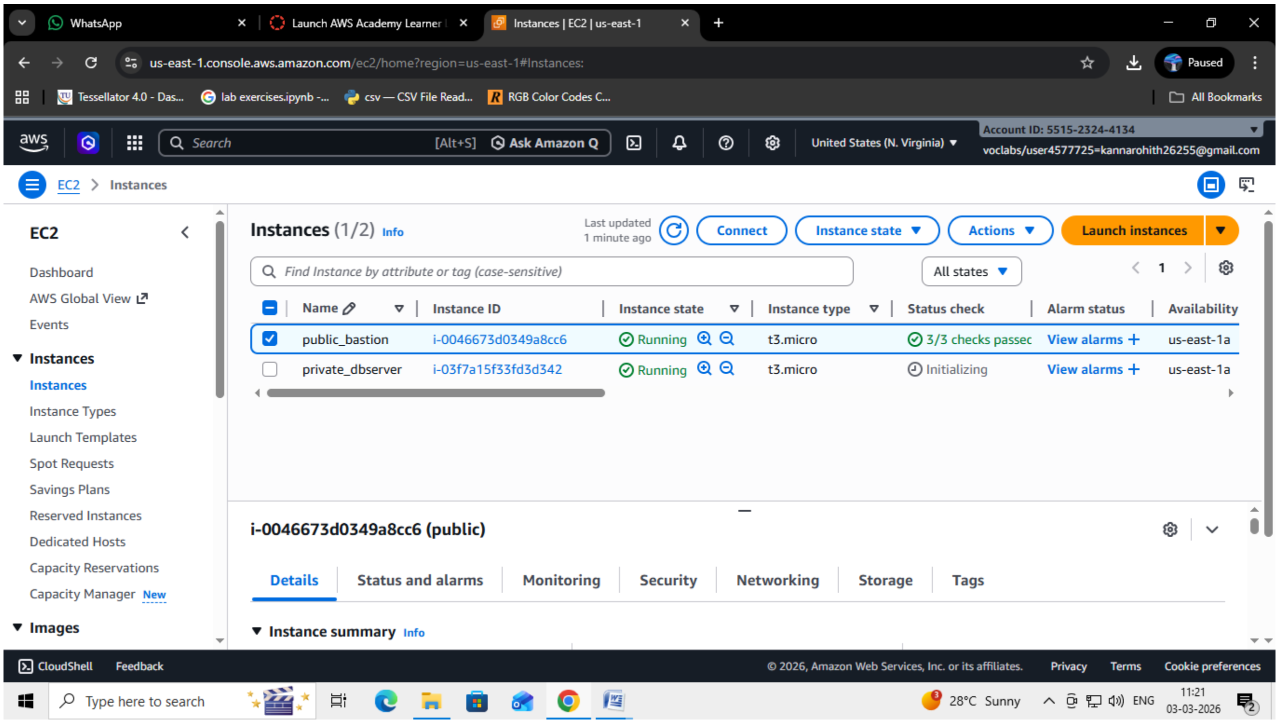




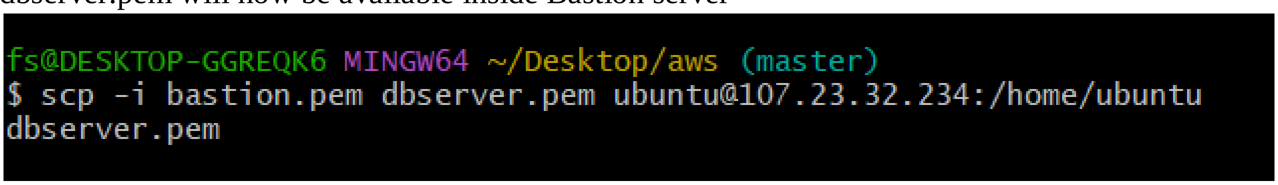
Launch an instance | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
EC2 > Instances > Launch an instance
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
dbserver
Create new key pair
Network settings Info
Edit
Network Info
vpc-08e6f3ebe3f57f6d6
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0b6c6ebed2801a5cb
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
Cancel
Launch instance
Preview code



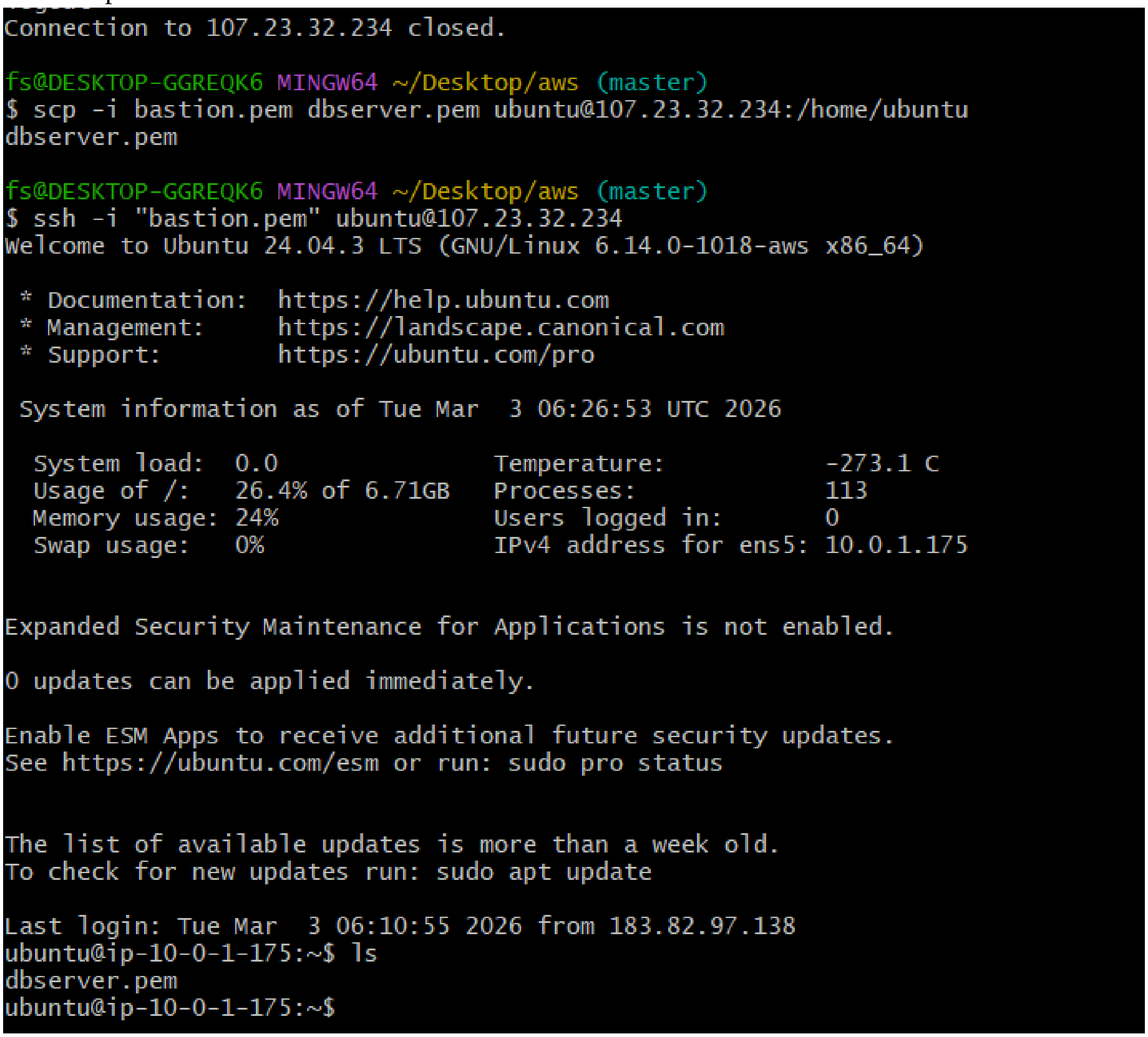
Launch an instance | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
EC2 > Instances > Launch an instance
VPC - required Info
vpc-0727f4d7d617191a6 (custom_vpc_exp6)
10.0.0.0/16
Subnet Info
subnet-0364e47f192a591ec private_subnet
VPC: vpc-0727f4d7d617191a6 Owner: 551523244134
Availability Zone: us-east-1a (use1-az6) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.2.0/24)
Create new subnet
Auto-assign public IP Info
Disable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
DatabaseSG sg-0d1043775e251f9a6
VPC: vpc-0727f4d7d617191a6
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0b6c6ebed2801a5cb
Virtual server type (instance type)
t3.micro
Firewall (security group)
DatabaseSG
Storage (volumes)
Cancel
Launch instance
Preview code



5. **Copy DB Server Key to Bastion Server**
Database server is in private subnet
Direct SSH from local system is NOT possible
DB server key must be copied to Bastion server
Command (PowerShell):
scp –i bastion.pem dbserver.pem ubuntu@<BASTION_PUBLIC_IP>:/home/ubuntu/
File is copied securely using SCP
dbserver.pem will now be available inside Bastion server



6. **Connect to Bastion Server**
   SSH Command:
   ssh -ibastion.pem ubuntu@<BASTION_PUBLIC_IP>
   Verify key files:
   ls
   Output:
   bastion.pem
   dbserver.pem

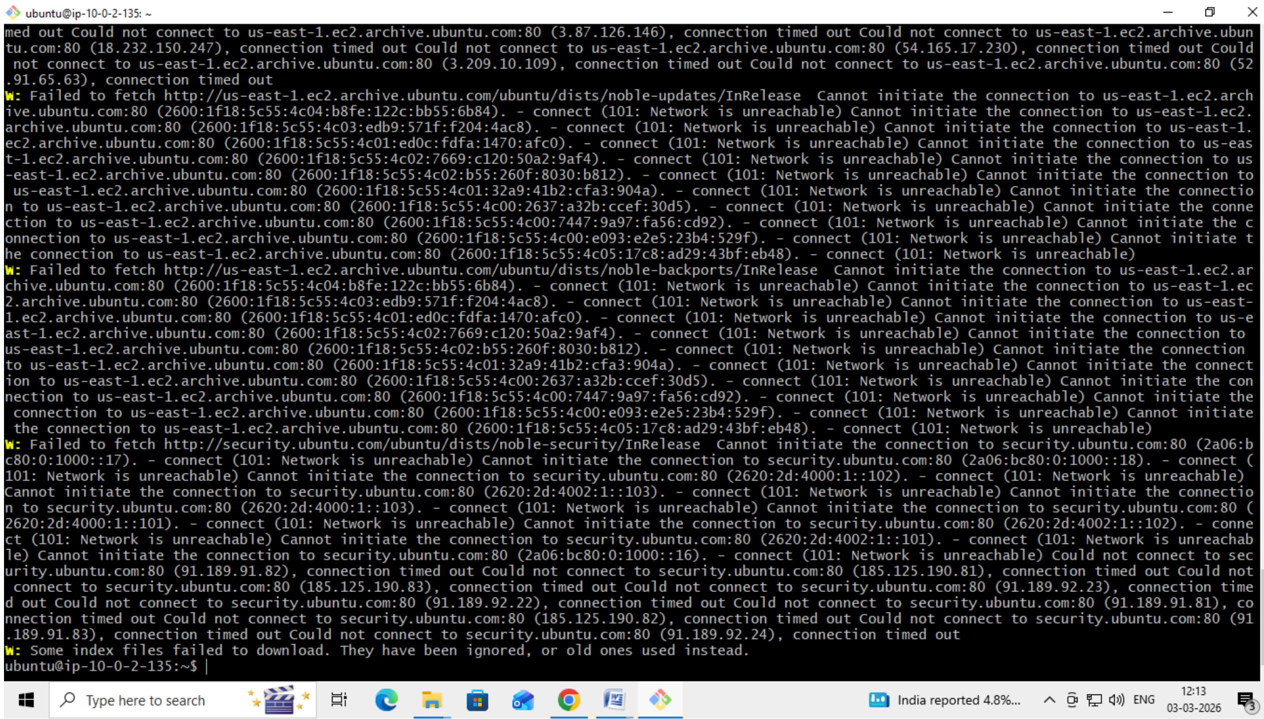


```
Connection to 107.23.32.234 closed.

fs@DESKTOP-GGREQK6 MINGW64 ~/Desktop/aws (master)
$ scp -i bastion.pem dbserver.pem ubuntu@107.23.32.234:/home/ubuntu
dbserver.pem

fs@DESKTOP-GGREQK6 MINGW64 ~/Desktop/aws (master)
$ ssh -i "bastion.pem" ubuntu@107.23.32.234
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Mar  3 06:26:53 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   26.4% of 6.71GB   Processes:             113
  Memory usage: 24%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 10.0.1.175


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Mar  3 06:10:55 2026 from 183.82.97.138
ubuntu@ip-10-0-1-175:~$ ls
dbserver.pem
ubuntu@ip-10-0-1-175:~$
```

7. **Connect from Bastion to Database Server**
   SSH from Bastion:
   ssh -idbserver.pem ubuntu@10.0.2.x
   Connection to DB server is successful
   Database administrator can now:
   Update database configurations
   Install required packages-------problem occurs

Restart MySQL service

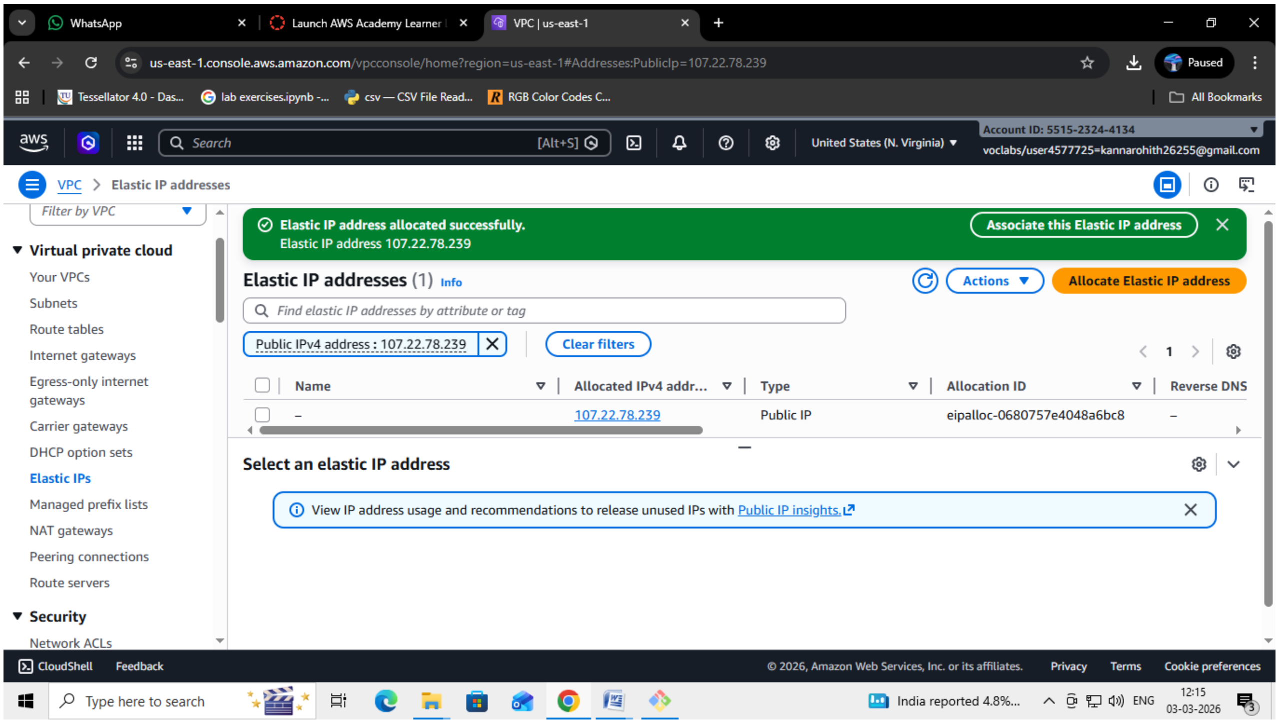
```
ubuntu@ip-10-0-1-175:~$ ls
dbserver.pem
ubuntu@ip-10-0-1-175:~$ ch mod 400 dbserver.pem
ch: command not found
ubuntu@ip-10-0-1-175:~$ chmod 400 dbserver.pem
ubuntu@ip-10-0-1-175:~$ ssh -i "dbserver.pem" ubuntu@10.0.2.135
The authenticity of host '10.0.2.135 (10.0.2.135)' can't be established.
ED25519 key fingerprint is SHA256:8sbYhPp+IaecCDEOsAsNTn9ayGmpCVk/RwOJj7CCJhQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.135' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Mar  3 06:34:06 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   26.4% of 6.71GB   Processes:             112
  Memory usage: 25%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 10.0.2.135

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
```



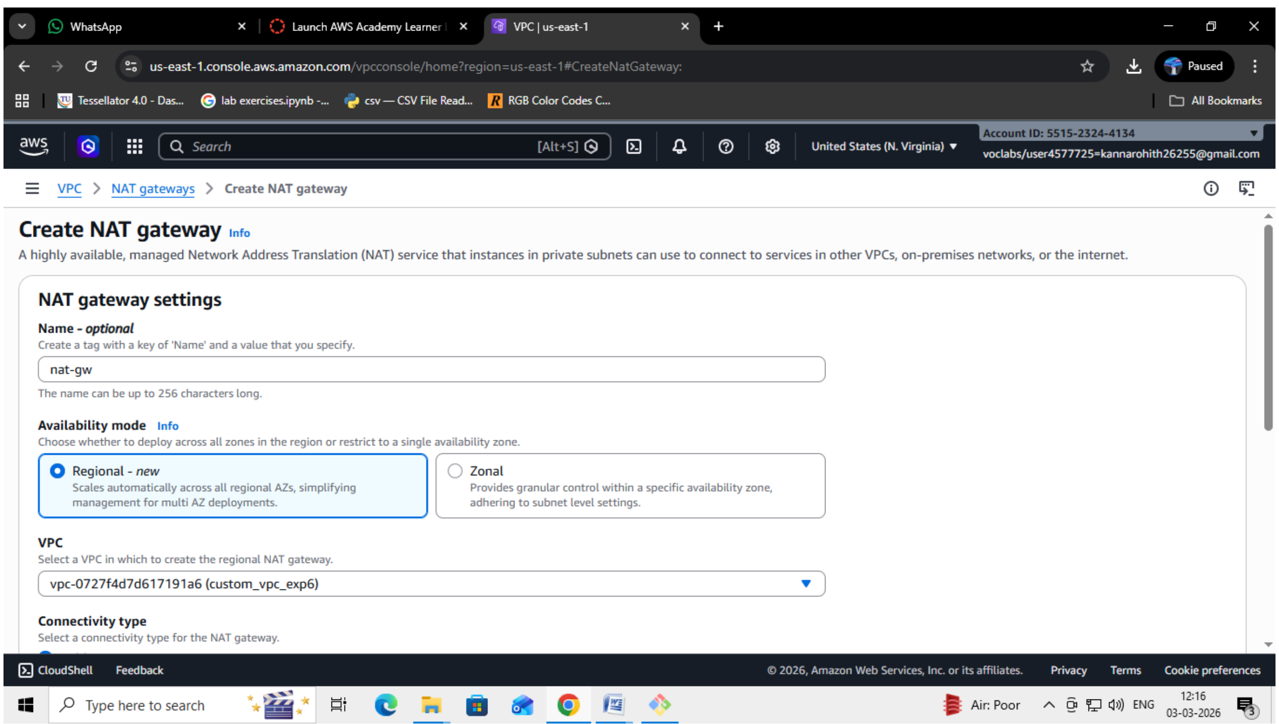
VPC USING NAT GATEWAY

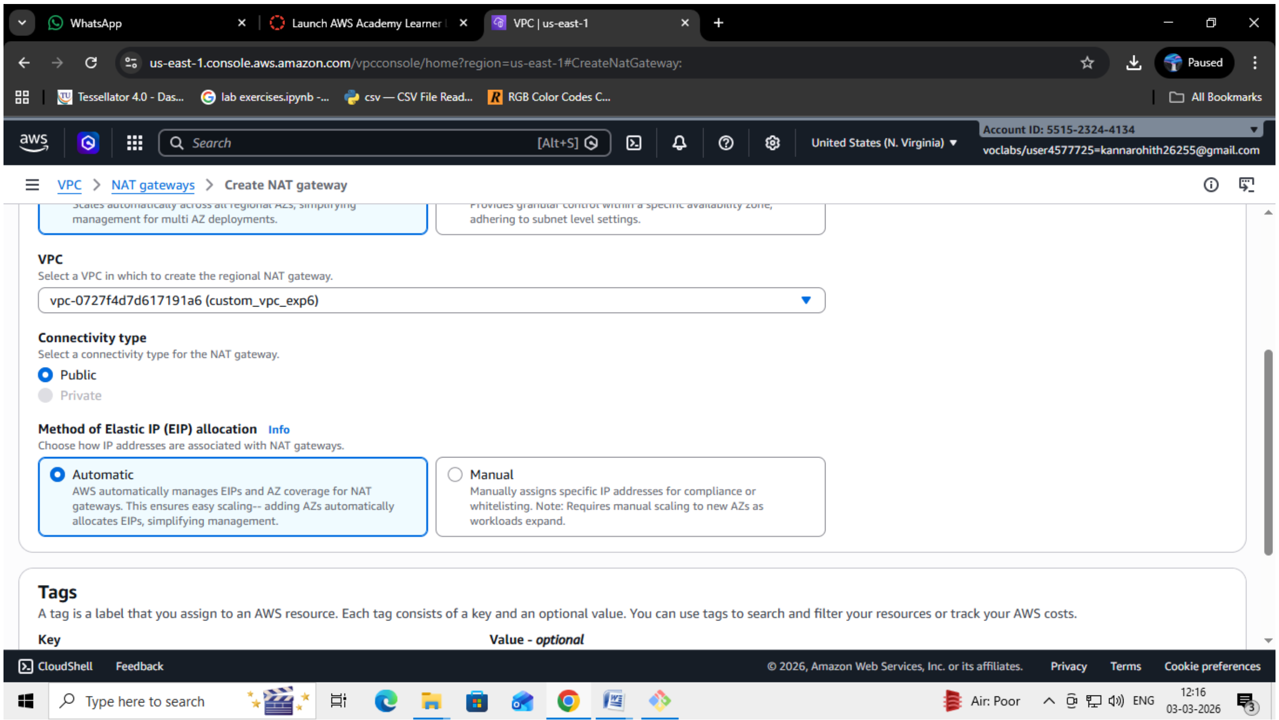
8. In the above steps it clear there is no internet access, if we want to install or update anything in the instances present in private subnet.
9. Network Address Translator (NAT Gateway) will solve this problem. Goto NAT Gateway in the left side menu bar of VPC and click on create NAT gateway.
10. Give the Nat gateway name, select your VPC, scroll down and click create NAT gateway.
11. So Now create a route for accessing NATGateway. Goto Route table click on create route table and give the route table name as bastion route , select our customized vpc and click on create route.
12. Select the route, click on actions, goto edit subnet associations and add private subnet to this bastion route.
13. Againgoto actions and click on edit routes, attach NAT Gateway with 0.0.0.0/0, select your NAT gateway name in the drop down menu and click on save changes.
14. Now check in the dbsever you will internet in ec2 of private subnet.

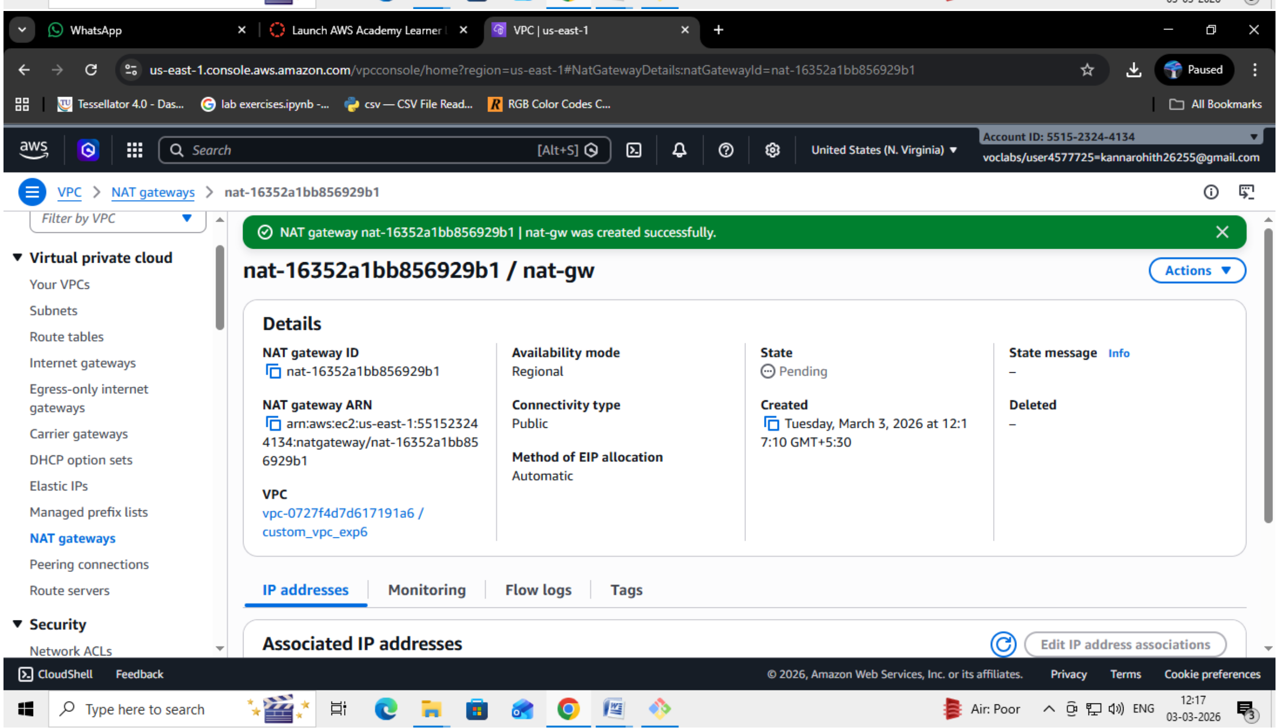
Allocate Elastic ip address:


WhatsApp
Launch AWS Academy Learner
VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#Addresses:PublicIp=107.22.78.239
Paused
Tessellator 4.0 - Das...
lab exercises.ipynb -...
csv — CSV File Read...
RGB Color Codes C...
All Bookmarks
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5515-2324-4134
voclabs/user4577725=kannarohith26255@gmail.com
VPC
Elastic IP addresses
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Elastic IP address allocated successfully.
Elastic IP address 107.22.78.239
Associate this Elastic IP address
Elastic IP addresses (1) Info
Actions
Allocate Elastic IP address
Find elastic IP addresses by attribute or tag
Public IPv4 address : 107.22.78.239
Clear filters
Name
Allocated IPv4 addr...
Type
Allocation ID
Reverse DNS
–
107.22.78.239
Public IP
eipalloc-0680757e4048a6bc8
–
Select an elastic IP address
View IP address usage and recommendations to release unused IPs with Public IP insights.
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
India reported 4.8%...
ENG
12:15
03-03-2026

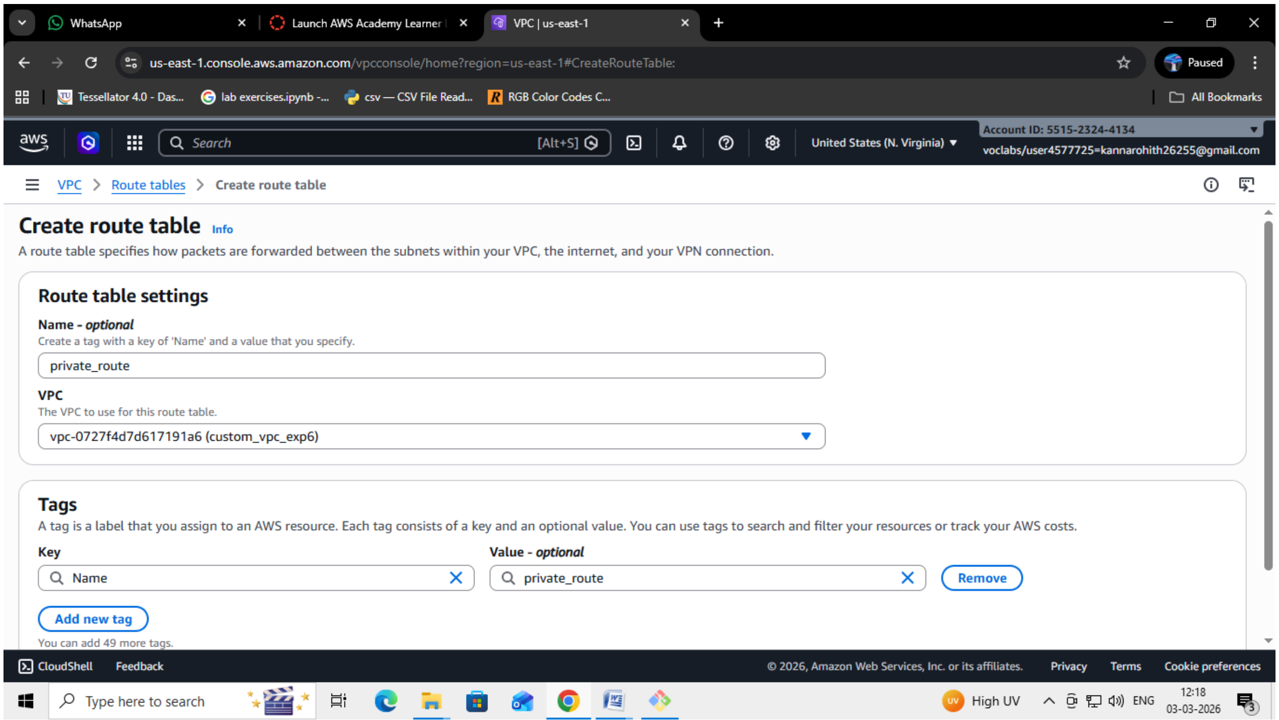
NAT GATEWAY:


WhatsApp
Launch AWS Academy Learner
VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateNatGateway:
Paused
Tessellator 4.0 - Das...
lab exercises.ipynb -...
csv — CSV File Read...
RGB Color Codes C...
All Bookmarks
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5515-2324-4134
voclabs/user4577725=kannarohith26255@gmail.com
VPC
NAT gateways
Create NAT gateway
Create NAT gateway Info
A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.
NAT gateway settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
nat-gw
The name can be up to 256 characters long.
Availability mode Info
Choose whether to deploy across all zones in the region or restrict to a single availability zone.
Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.
Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.
VPC
Select a VPC in which to create the regional NAT gateway.
vpc-0727f4d7d617191a6 (custom_vpc_exp6)
Connectivity type
Select a connectivity type for the NAT gateway.
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
Air: Poor
ENG
12:16
03-03-2026

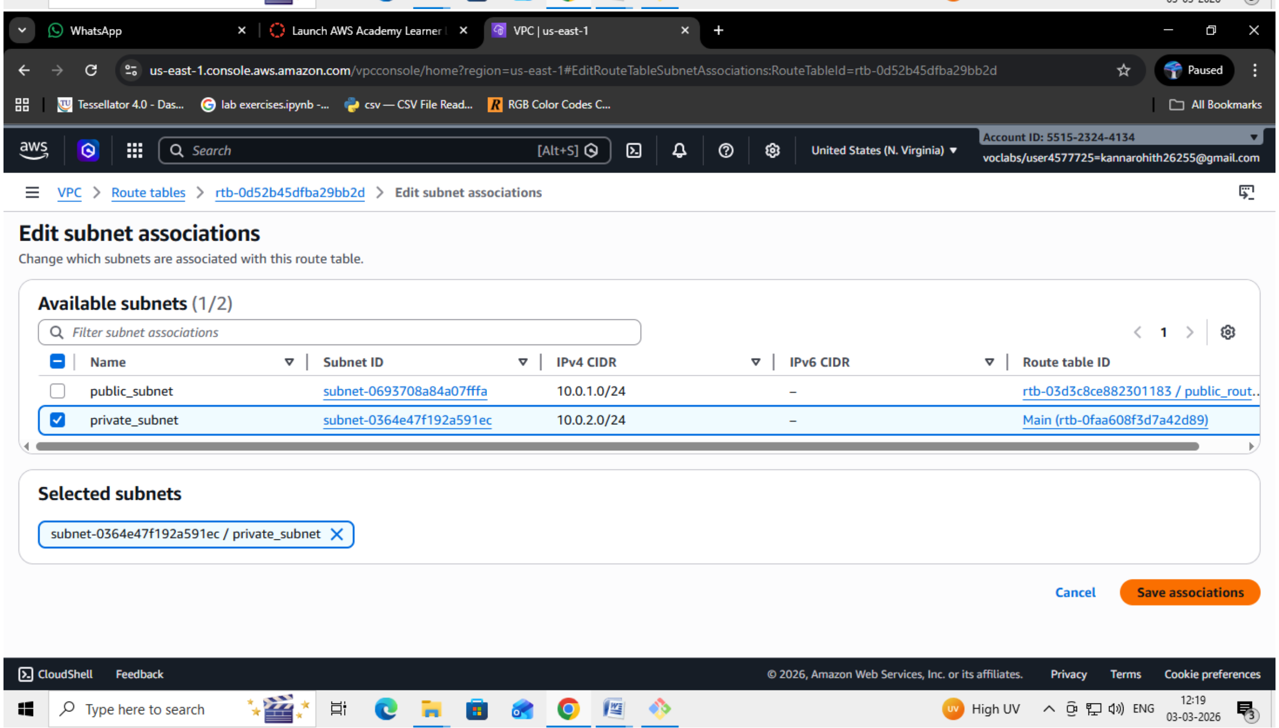

WhatsApp
Launch AWS Academy Learner
VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateNatGateway:
Paused
Tessellator 4.0 - Das...
lab exercises.ipynb -...
csv — CSV File Read...
RGB Color Codes C...
All Bookmarks
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5515-2324-4134
voclabs/user4577725=kannarohith26255@gmail.com
VPC
NAT gateways
Create NAT gateway
management for multi AZ deployments.
adhering to subnet level settings.
VPC
Select a VPC in which to create the regional NAT gateway.
vpc-0727f4d7d617191a6 (custom_vpc_exp6)
Connectivity type
Select a connectivity type for the NAT gateway.
Public
Private
Method of Elastic IP (EIP) allocation
Info
Choose how IP addresses are associated with NAT gateways.
Automatic
AWS automatically manages EIPs and AZ coverage for NAT gateways. This ensures easy scaling-- adding AZs automatically allocates EIPs, simplifying management.
Manual
Manually assigns specific IP addresses for compliance or whitelisting. Note: Requires manual scaling to new AZs as workloads expand.
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Value - optional
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
Air: Poor
ENG
12:16
03-03-2026


WhatsApp
Launch AWS Academy Learner
VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#NatGatewayDetails:natGatewayId=nat-16352a1bb856929b1
Paused
Tessellator 4.0 - Das...
lab exercises.ipynb -...
csv — CSV File Read...
RGB Color Codes C...
All Bookmarks
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5515-2324-4134
voclabs/user4577725=kannarohith26255@gmail.com
VPC
NAT gateways
nat-16352a1bb856929b1
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
NAT gateway nat-16352a1bb856929b1 | nat-gw was created successfully.
nat-16352a1bb856929b1 / nat-gw
Actions
Details
NAT gateway ID
nat-16352a1bb856929b1
NAT gateway ARN
arn:aws:ec2:us-east-1:551523244134:natgateway/nat-16352a1bb856929b1
VPC
vpc-0727f4d7d617191a6 / custom_vpc_exp6
Availability mode
Regional
Connectivity type
Public
Method of EIP allocation
Automatic
State
Pending
Created
Tuesday, March 3, 2026 at 12:17:10 GMT+5:30
State message
Info
–
Deleted
–
IP addresses
Monitoring
Flow logs
Tags
Associated IP addresses
Edit IP address associations
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
Air: Poor
ENG
12:17
03-03-2026

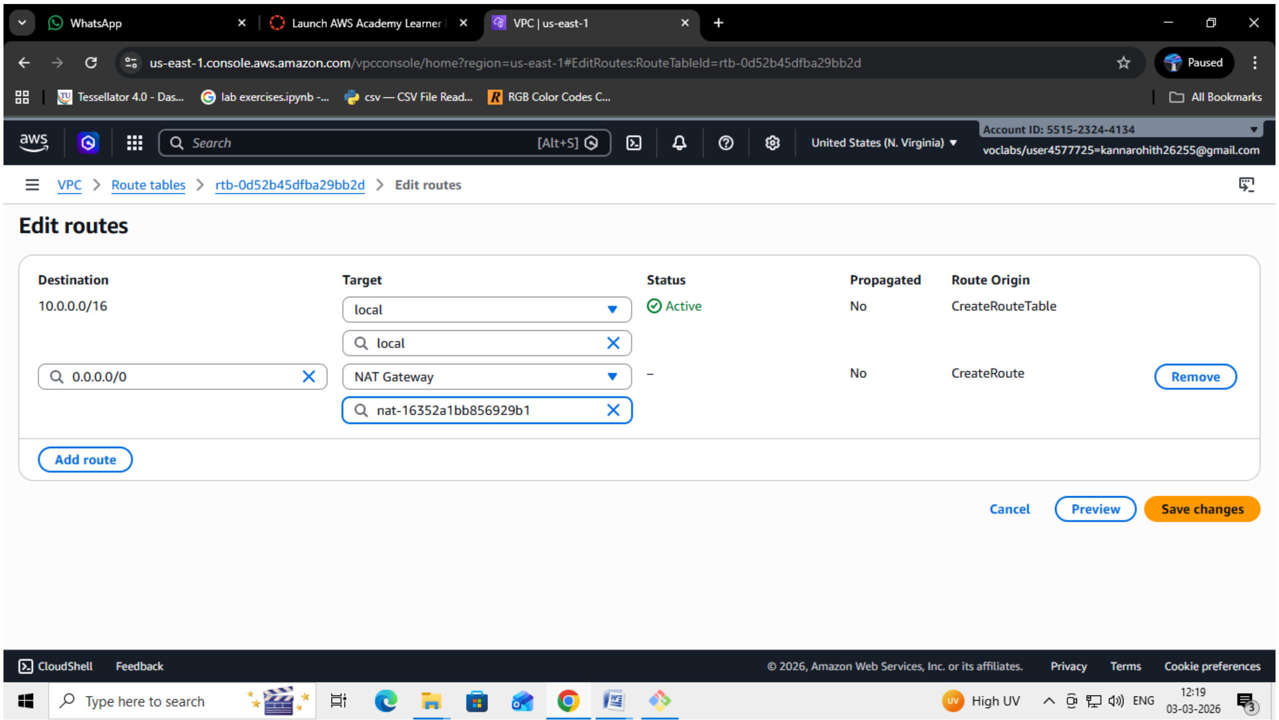
Route_table:



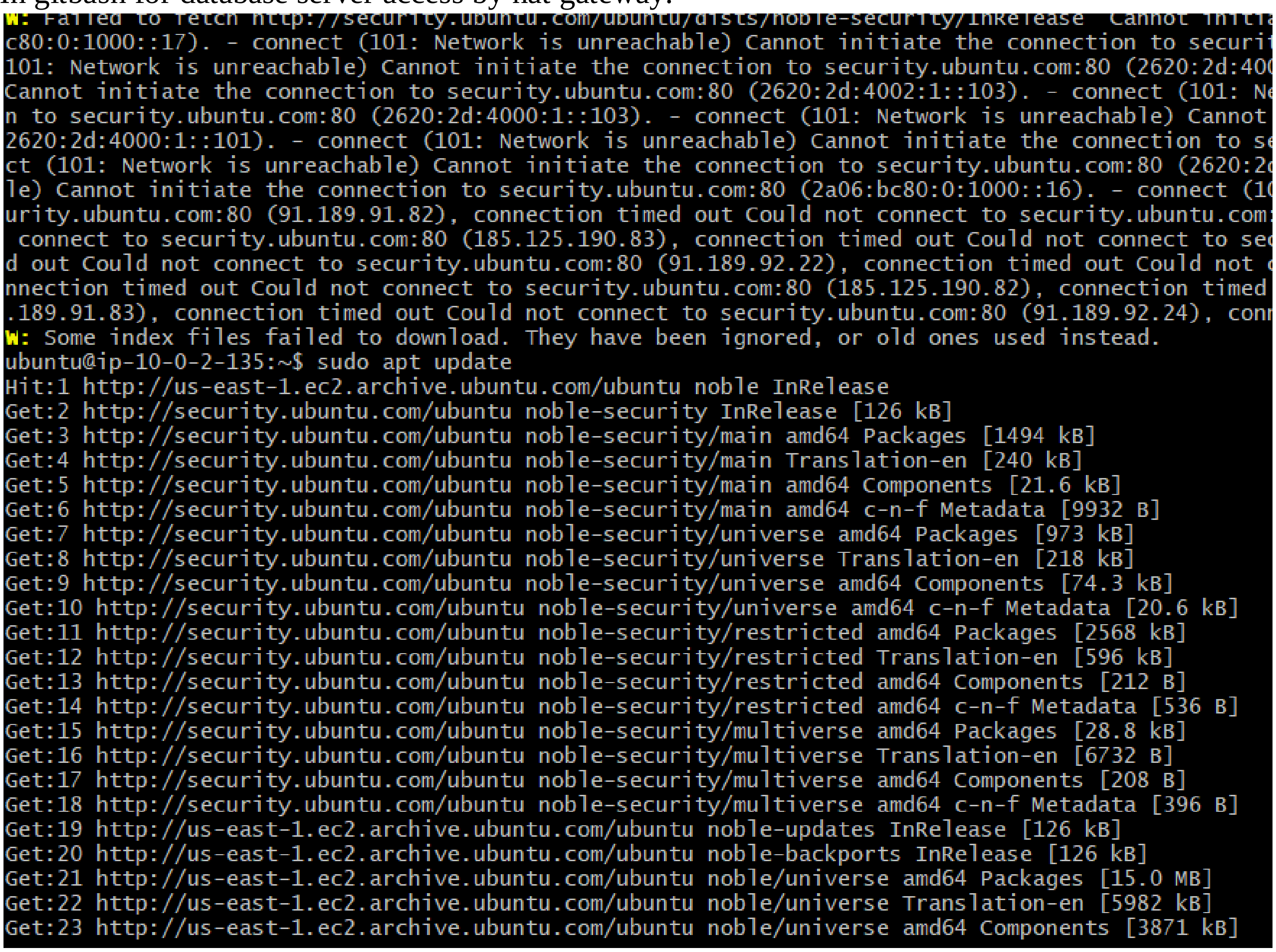
WhatsApp
Launch AWS Academy Learner
VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateRouteTable:
Paused
Tessellator 4.0 - Das...
lab exercises.ipynb -...
csv — CSV File Read...
RGB Color Codes C...
All Bookmarks
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5515-2324-4134
voclabs/user4577725=kannarohith26255@gmail.com
VPC > Route tables > Create route table
Create route table
Info
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
private_route
VPC
The VPC to use for this route table.
vpc-0727f4d7d617191a6 (custom_vpc_exp6)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Name
Value - optional
private_route
Remove
Add new tag
You can add 49 more tags.
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
High UV
ENG
12:18
03-03-2026

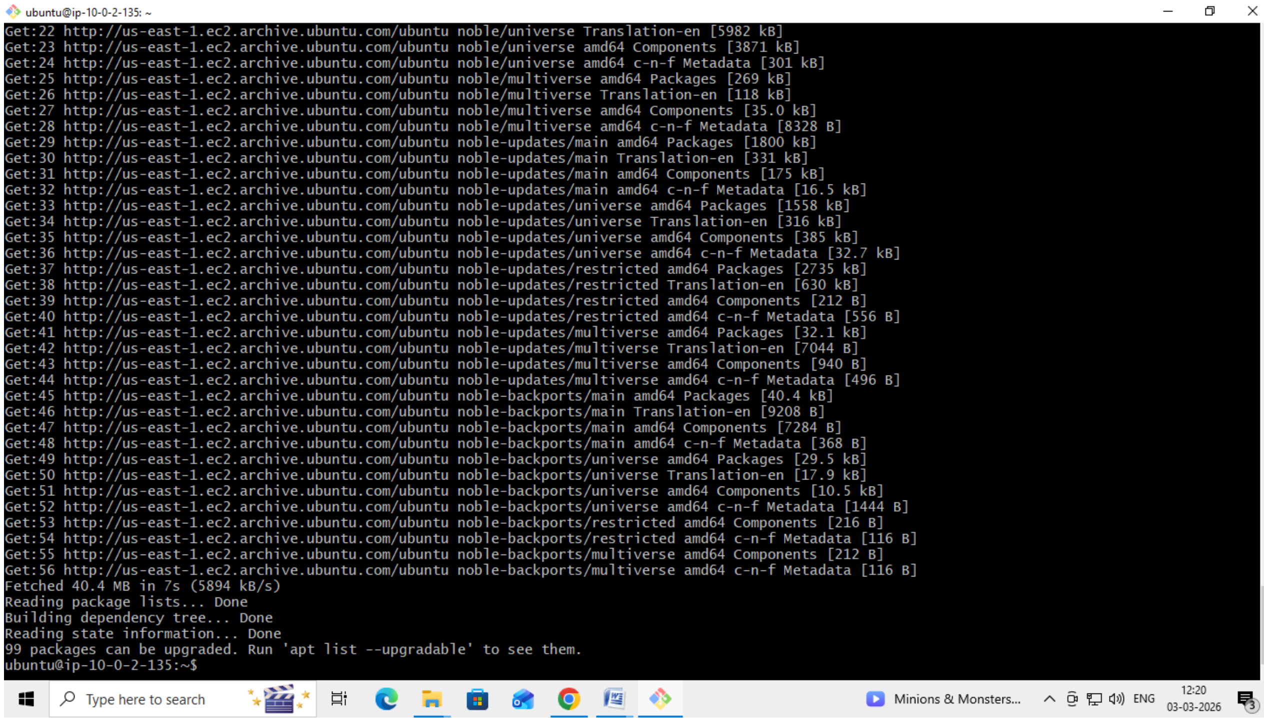


WhatsApp
Launch AWS Academy Learner
VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditRouteTableSubnetAssociations:RouteTableId=rtb-0d52b45dfba29bb2d
Paused
Tessellator 4.0 - Das...
lab exercises.ipynb -...
csv — CSV File Read...
RGB Color Codes C...
All Bookmarks
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5515-2324-4134
voclabs/user4577725=kannarohith26255@gmail.com
VPC > Route tables > rtb-0d52b45dfba29bb2d > Edit subnet associations
Edit subnet associations
Change which subnets are associated with this route table.
Available subnets (1/2)
Filter subnet associations
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
Route table ID
public_subnet
subnet-0693708a84a07fffa
10.0.1.0/24
–
rtb-03d3c8ce882301183 / public_rout...
private_subnet
subnet-0364e47f192a591ec
10.0.2.0/24
–
Main (rtb-0faa608f3d7a42d89)
Selected subnets
subnet-0364e47f192a591ec / private_subnet
Cancel
Save associations
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
High UV
ENG
12:19
03-03-2026



In gitbash for database server access by nat gateway:





SCENARIO BASED QUESTIONS

1. You have multiple private subnets in different AZs. Should you use **one NAT Gateway or multiple NAT Gateways**? Explain your design choice.
2. Your private subnet instances can reach the internet, but the NAT Gateway shows no traffic. What could be the possible reasons?
3. A NAT Gateway was accidentally created in a **private subnet** instead of a public subnet. What prob lem will occur, and how would you fix it?
4. An application running in a private subnet requires outbound access only to a few trusted external services.
   How would you restrict traffic while still using a NAT Gateway?
5. Multiple teams share a single VPC and NAT Gateway. One team's traffic is affecting others. How would you redesign the NAT Gateway setup to isolate traffic?
6. You can successfully SSH into a private EC2 instance using a Bastion server, but yum update / apt update fails inside the private instance. Why?
7. Private EC2 instances can access the internet through a NAT Gateway, but administrators cannot SSH into them using the Bastion server.
8. A Bastion server and NAT Gateway are both created in a **private subnet** by mistake. What impact does this cause?
9. SSH access to the Bastion server works, but SSH from Bastion to the private EC2 fails. Why?
10. A student claims that a Bastion server alone is enough to provide internet access to private instances. Why is Bastion not a replacement for NAT Gateway?